

MedDRA Coding Practice Project

Adverse Event Term Mapping to the MedDRA Hierarchy | Self-Directed Pharmacovigilance Training
Nishanth P | Dharmapuri, India | Spnishanth18@gmail.com | linkedin.com/in/nishanth-poovarasan

OBJECTIVE

To build hands-on proficiency in MedDRA (Medical Dictionary for Regulatory Activities) coding — a core skill in ICSR case processing — by taking verbatim adverse event descriptions from illustrative case narratives and mapping each one through the MedDRA hierarchy (Lowest Level Term → Preferred Term → High Level Term → High Level Group Term → System Organ Class), following WHO-UMC and MedDRA coding conventions used in pharmacovigilance practice.

BACKGROUND

Built as a follow-on to a self-directed ICSR narrative writing exercise, this project focuses specifically on the coding step of case processing: translating free-text adverse event descriptions into standardized, queryable MedDRA terms. Ten verbatim terms were selected to span common pharmacovigilance scenarios — hepatobiliary, dermatological, cardiac, gastrointestinal, neurological, psychiatric, and medication-error events — reflecting the variety of terms a case processor encounters day to day.

METHODOLOGY

- Selected verbatim AE terms from illustrative case narratives, written the way a reporter or clinician would naturally phrase them (not pre-coded).
- Identified the most specific matching Lowest Level Term (LLT) for each verbatim, avoiding vaguer 'not otherwise specified' terms where a more precise LLT existed.
- Mapped each LLT to its linked Preferred Term (PT), then traced the PT up through its High Level Term (HLT), High Level Group Term (HLGT), and primary System Organ Class (SOC).
- Applied core MedDRA conventions: one PT per distinct medical concept, precoordination of terms where applicable, and awareness of multi-axial SOC linkages for terms that logically sit under more than one SOC.
- Cross-checked coding logic against MedDRA Points to Consider guidance and WHO-UMC causality/coding principles referenced during certification study.
- Documented the rationale for each coding decision in a structured worksheet, mirroring the audit-trail discipline expected in SOP-driven case processing environments.

SAMPLE CODING WORKSHEET

Illustrative mapping of ten verbatim adverse event terms through the MedDRA hierarchy:

#	Verbatim Term (Reported)	LLT	PT	HLT	HLGT	SOC
1	Yellowing of the eyes and skin	Jaundice	Jaundice	Hepatobiliary signs and symptoms	Hepatobiliary disorders NEC	Hepatobiliary disorders
2	Severe itching and hives all over the body	Urticaria	Urticaria	Urticarias	Epidermal and dermal conditions	Skin and subcutaneous tissue disorders
3	Nurse gave the wrong insulin dose	Incorrect dose administered	Incorrect dose administered	Medication errors	Medication errors NEC	Injury, poisoning and procedural complications
4	Lips and throat swelling, trouble breathing	Angioedema	Angioedema	Angioedema and urticaria	Allergic conditions	Immune system disorders

#	Verbatim Term (Reported)	LLT	PT	HLT	HLGT	SOC
5	Elevated liver enzymes on routine bloodwork	Hepatic enzyme increased	Hepatic enzyme increased	Liver function analyses	Hepatobiliary and pancreatic investigations	Investigations
6	Severe headache and dizziness after first dose	Headache	Headache	Headaches NEC	Neurological disorders NEC	Nervous system disorders
7	Irregular heartbeat noted on ECG	Arrhythmia	Arrhythmia	Cardiac arrhythmias NEC	Cardiac arrhythmias	Cardiac disorders
8	Nausea and vomiting after taking the tablet	Vomiting	Vomiting	Gastrointestinal signs and symptoms NEC	Gastrointestinal motility and defaecation conditions	Gastrointestinal disorders
9	Patient developed a rash and fever	Rash	Rash	Rashes, eruptions and exanthems NEC	Epidermal and dermal conditions	Skin and subcutaneous tissue disorders
10	Trouble sleeping since starting the medication	Insomnia	Insomnia	Sleep disorders and disturbances	Sleep disorders	Psychiatric disorders

Note: term mappings are illustrative and constructed for self-training purposes to reflect standard MedDRA hierarchy conventions. They are not reproduced from a licensed MedDRA browser and should be re-verified against the current MedDRA version before any professional or regulatory use.

SKILLS DEMONSTRATED

- MedDRA multi-axial hierarchy navigation (LLT / PT / HLT / HLGT / SOC)
- Selection of the most specific, medically accurate coding term from a verbatim description
- Application of WHO-UMC and MedDRA “Points to Consider” coding conventions
- Structured documentation of coding rationale for audit readiness and QC review
- Recognition of coding scenarios relevant to SUSAR/PSUR/DSUR-supporting case data

TOOLS & REFERENCES

MedDRA hierarchy structure and coding conventions; WHO-UMC coding and causality assessment principles; MS Excel (coding worksheet); concepts reinforced through MedDRA Training Certificate coursework and Advanced Pharmacovigilance and Drug Safety certification.

Self-directed project completed to build coding readiness ahead of CRO-side ICSR processing and pharmacovigilance case management roles.